

# folland 1

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August 2026

## 1 $\sigma$ -Algebras

Claim: Let  $\{(X_i, d_i)\}_{i \in [n]}$  be metric spaces and  $X = \prod_{i \in [n]} X_i$ . Then  $\bigotimes_{i \in [n]} \mathcal{B}_{X_i} \leq \mathcal{B}_X$ . If the  $X_i$ 's are separable, then  $\bigotimes_{i \in [n]} \mathcal{B}_{X_i} = \mathcal{B}_X$ .

Proof:  $\bigotimes_{i \in [n]} \mathcal{B}_{X_i}$  is generated by the sets  $\prod_{i \in [n]} U_i$ , where  $U_i \leq X_i$  is open. The product of open sets is open, so  $\bigotimes_{i \in [n]} \mathcal{B}_{X_i} \leq \mathcal{B}_X$ . Suppose the  $X_i$  are separable. Let  $D_i$  be a countable dense subset in  $X_i$  and let  $U \leq X$  be open. Then  $D = \prod_{i \in [n]} D_i$  is countable dense in  $X$ . Fix  $k \in \mathbb{Z}^+$  and define  $\mathcal{V}_k = \{V_x = \prod_{i \in [n]} B_{d_i}(x_i, 1/k) : x \in U \cap D, V_x \leq U\}$ . I claim that  $\mathcal{V} = \bigcup_{k \in \mathbb{Z}^+} \mathcal{V}_k$  is a countable open cover of  $\mathcal{V}$ .

Each  $\mathcal{V}_k$  is countable, so  $\mathcal{V}$  is countable. We have  $\bigcup \mathcal{V} \leq U$ . Suppose  $x \in U$ . Pick  $k \in \mathbb{Z}^+$  such that  $\prod_{i \in [n]} B_{d_i}(x_i, 1/k) \leq U$ . Since  $D_i$  is dense in  $X_i$ , pick  $y_i \in B_{d_i}(x_i, 1/(2k)) \cap D_i$ . Let  $y = (y_i)_{i \in [n]}$ . I claim that  $x \in V_y = \prod_{i \in [n]} B_{d_i}(y_i, 1/(2k)) \leq \prod_{i \in [n]} B_{d_i}(x_i, 1/k)$ . The membership follows from the symmetry of  $d_i$ . Suppose  $z \in \prod_{i \in [n]} B_{d_i}(y_i, 1/(2k))$ . Then  $d_i(z_i, x_i) \leq d_i(z_i, y_i) + d_i(y_i, x_i) < 1/2k + 1/2k = 1/k$ . Thus,  $x \in V_y \in \mathcal{V}_{2k}$ .

Claim: Let  $\mathcal{E}$  be an elementary family. Then the collection  $\mathcal{A}$  of finite disjoint unions of elements of  $\mathcal{E}$  is an algebra.

Proof: Let  $\{E_i\}_{i \in [n]}$  in  $\mathcal{E}$  be pairwise disjoint. Let  $E_i^c = \bigsqcup_{j_i \in [n_i]} E_{i,j_i}$ . Then

$$\left( \bigsqcup_{i \in [n]} E_i \right)^c = \bigcap_{i \in [n]} E_i^c = \bigcap_{i \in [n]} \bigsqcup_{j_i \in [n_i]} E_{i,j_i} = \bigsqcup_{j \in \prod_{i \in [n]} [n_i]} \bigcap_{i \in [n]} E_{i,j_i}.$$

Note that  $\bigcap_{i \in [n]} E_{i,j_i} \in \mathcal{E}$  and that these sets are disjoint for distinct  $j$ , so the above set is in  $\mathcal{A}$ .

Let  $\{F_j\}_{j \in [m]}$  be another collection in  $\mathcal{E}$ . Then

$$\left( \bigsqcup_{i \in [n]} E_i \right) \cap \left( \bigsqcup_{j \in [m]} F_j \right) = \bigsqcup_{(i,j) \in [n] \times [m]} E_i \cap F_j,$$

which is a disjoint union of elements in  $\mathcal{E}$ .

## 2 Measures

Claim (completion of measure): Let  $(X, \mathcal{M}, \mu)$  be a measure. Define  $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$  and  $\overline{\mathcal{M}} = \{M \cup F : M \in \mathcal{M}, F \leq N \text{ for some } N \in \mathcal{N}\}$ . Then  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra and  $\mu$  extends uniquely to the complete measure  $\bar{\mu}$  on  $\overline{\mathcal{M}}$ .

Proof:  $\emptyset \in \overline{\mathcal{M}}$ . Let  $M \cup F \in \overline{\mathcal{M}}$ . Then

$$\begin{aligned} (M \cup F)^c &= M^c \cap F^c = M^c \cup N^c \cup (F^c - N^c) \\ &= M^c \cup N^c \cup (N - F) \in \overline{\mathcal{M}} \end{aligned}$$

since  $N - F \leq N$  and  $M^c \cup N^c \in \mathcal{M}$ . Let  $\{M_n \cup F_n\}_{n \in \mathbb{Z}^+}$  be in  $\overline{\mathcal{M}}$  and let  $N_n$  denote the corresponding  $N$ 's. Then

$$\bigcup_{n \in \mathbb{Z}^+} M_n \cup F_n = \left( \bigcup_{n \in \mathbb{Z}^+} M_n \right) \cup \left( \bigcup_{n \in \mathbb{Z}^+} F_n \right) \in \overline{\mathcal{M}}$$

since  $\bigcup_{n \in \mathbb{Z}^+} M_n \in \mathcal{M}$  and  $\bigcup_{n \in \mathbb{Z}^+} F_n \leq \bigcup_{n \in \mathbb{Z}^+} N_n$ . Note  $\mu(\bigcup_{n \in \mathbb{Z}^+} N_n) \leq \sum_{n \in \mathbb{Z}^+} \mu(N_n) = 0$ . Thus,  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra.

Define  $\bar{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$  by  $\bar{\mu}(M \cup F) = \mu(M)$ . I claim that  $\bar{\mu}$  is well-defined. Suppose  $M \cup F = M' \cup F'$ . Then

$$M \cup N \cup N' = (M \cup F) \cup (N \cup N') = (M' \cup F') \cup (N \cup N') = M' \cup N \cup N'$$

Thus,

$$\mu(M) \leq \mu(M \cup N \cup N') = \mu(M' \cup N \cup N') \leq \mu(M') + \mu(N \cup N') = \mu(M').$$

Similarly,  $\mu(M') \leq \mu(M)$ , so  $\mu(M) = \mu(M')$ .

We have  $\bar{\mu}(\emptyset) = \mu(\emptyset) = 0$ . Let  $\{M_n \cup F_n\}_{n \in \mathbb{Z}^+}$  in  $\overline{\mathcal{M}}$  be disjoint. Then

$$\bar{\mu}\left(\bigcup_{n \in \mathbb{Z}^+} M_n \cup F_n\right) = \mu\left(\bigcup_{n \in \mathbb{Z}^+} M_n\right) = \sum_{n \in \mathbb{Z}^+} \mu(M_n) = \sum_{n \in \mathbb{Z}^+} \bar{\mu}(M_n \cup F_n),$$

showing additivity.

I claim that  $\bar{\mu}$  is unique. Suppose  $\bar{\mu}$  is a measure on  $\overline{\mathcal{M}}$  extending  $\mu$ . Then:

$$\mu(M) = \bar{\mu}(M) \leq \bar{\mu}(M \cup F) \leq \bar{\mu}(M \cup N) = \mu(M \cup N) \leq \mu(M)$$

Finally, I show that  $\bar{\mu}$  is complete. Suppose  $\bar{\mu}(M \cup F) = \mu(M) = 0$ . Let  $M' \leq M \cup F$ , so  $M' \leq M \cup N$ , so  $M' \in \overline{\mathcal{M}}$ .

### 3 Outer Measures

Claim: Let  $\mathcal{E} \leq \mathcal{P}(X)$  be such that  $\emptyset, X \in \mathcal{E}$  and let  $\rho : \mathcal{E} \rightarrow [0, \infty]$ . Define

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \rho(E_n) : A \leq \bigcup_{n \in \mathbb{Z}^+} E_n \right\}$$

Then  $\mu^*$  is an outer measure.

Proof: Note that  $\mu^*$  is well-defined since we can cover any  $A$  with  $X$ .  $\mu^*$  is nonnegative since  $\rho$  is nonnegative. We can cover  $\emptyset$  with  $\emptyset \in \mathcal{E}$ , so  $\mu^*(\emptyset) = 0$ . Suppose  $A \leq B$ . We wish to show  $\mu^*(A) \leq \mu^*(B)$ . Indeed,

$$\left\{ \sum_{n=1}^{\infty} \rho(E_n) : A \leq \bigcup_{n \in \mathbb{Z}^+} E_n \right\} \geq \left\{ \sum_{n=1}^{\infty} \rho(E_n) : B \leq \bigcup_{n \in \mathbb{Z}^+} E_n \right\}$$

since a cover of  $B$  is also a cover of  $A$ .

Let  $\epsilon > 0$ . For each  $i \in \mathbb{Z}^+$ , pick a cover  $\{E_{ij}\}_{j \in \mathbb{Z}^+}$  of  $A_i$  such that  $\sum_{j=1}^{\infty} \rho(E_{ij}) \leq \mu^*(A_i) + \epsilon/2^i$ . Note that  $\{E_{ij}\}_{i,j \in \mathbb{Z}^+}$  is a cover of  $\bigcup_{i \in \mathbb{Z}^+} A_i$ . Then

$$\sum_{i,j \in \mathbb{Z}^+} \rho(E_{ij}) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \rho(E_{ij}) \leq \sum_{i=1}^{\infty} \mu^*(A_i) + \epsilon/2^i = \epsilon + \sum_{i=1}^{\infty} \mu^*(A_i)$$

The LHS is at least  $\mu^*(\bigcup_{n \in \mathbb{Z}^+} A_n)$ . Since this holds for all  $\epsilon > 0$ , we have

$$\mu^*\left(\bigcup_{n \in \mathbb{Z}^+} A_n\right) \leq \sum_{n \in \mathbb{Z}^+} \mu^*(A_n),$$

as desired.

Claim (Caratheodory's theorem): If  $\mu^*$  is an outer measure on  $X$ , then the set of  $\mu^*$ -measurable functions  $\mathcal{M}$  is a  $\sigma$ -algebra, and the restriction of  $\mu^*$  to  $\mathcal{M}$  is a complete measure.

Proof:  $\emptyset$  is  $\mu^*$ -measurable and if  $A$  is  $\mu^*$ -measurable, so is  $A^c$ . We first show that  $\mathcal{M}$  is an algebra. Let  $A, B$  be  $\mu^*$ -measurable. Then

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) \\ &\geq \mu^*(E \cap A \cap B) + \mu^*(E \cap (A \cap B)^c), \end{aligned}$$

so  $A \cup B$  is  $\mu^*$ -measurable.

Let  $\{A_n\}_{n \in \mathbb{Z}^+}$  be disjoint  $\mu^*$ -measurable and denote by  $A = \bigcup_{n \in \mathbb{Z}^+} A_n$ . Then

$$\begin{aligned} \mu^*(E \cap A) + \mu^*(E \cap A^c) &\leq \lim_{n \rightarrow \infty} \left( \sum_{k=1}^n \mu^*(E \cap A_k) + \mu^*(E \cap A_k^c) \right) \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \mu^*(E) = \mu^*(E) \end{aligned}$$

so  $A$  is  $\mu^*$ -measurable. Countable additivity is satisfied.  $\mu^*$  is complete.

Claim: Let  $\mu_0$  be a premeasure on  $\mathcal{A} \leq \mathcal{P}(X)$ . Let  $\mu^*$  be the induced outer measure. Then  $\mu^*|_{\mathcal{A}} = \mu_0$  and every set in  $\mathcal{A}$  is  $\mu^*$ -measurable.

Proof: Suppose  $A \in \mathcal{A}$  and let  $\{A_n\}$  be a cover. Denote by  $B_n = \bigcup_{k \in [n]} A_k$ . Then

$$\sum_{n=1}^{\infty} \mu_0(A_n) \geq \sum_{n=1}^{\infty} \mu_0(B_n \cap A) = \mu_0(A).$$

$A$  can be covered by itself, so  $\mu^*(A) = \mu_0(A)$ .

Let  $\{A_n\}$  be a cover of  $E$ . Let  $B_n = A_n \cap A$  and  $C_n = A_n \cap A^c$ , so  $\{B_n\}, \{C_n\}$  are covers of  $E \cap A, E \cap A^c$ , respectively. Then

$$\begin{aligned} \sum_{n \in \mathbb{Z}^+} \mu_0(A_n) &= \sum_{n \in \mathbb{Z}^+} \mu_0(B_n) + \mu_0(C_n) = \sum_{n \in \mathbb{Z}^+} \mu_0(B_n) + \sum_{n \in \mathbb{Z}^+} \mu_0(C_n) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c). \end{aligned}$$

The cover  $\{A_n\}$  is arbitrary, so  $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$ .